TWITTER FINANCIAL NEWS ANALYSIS

Technologies
Python • NLP • Deep Learning • TensorFlow • Scikit-learn

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BUSINESS PROBLEM

Financial markets generate massive volumes of textual information from:

- Twitter
- Financial news
- Analyst updates
- Market commentary

Challenges:

- Difficult to monitor financial information in real time
- Manual classification of news topics is inefficient
- Analysts need faster insights from textual financial data

Business Question

How can we automatically classify financial tweets into meaningful financial topics?

PROJECT OBJECTIVES

1. Build an automated NLP pipeline
2. Analyze financial tweet dataset
3. Transform text data into machine learning features
4. Train deep learning models for classification
5. Evaluate model performance

BUSINESS IMPACT

(WHY THIS PROJECT MATTERS??)

Automated financial text classification enables:

- Faster financial news analysis
- Improved investment intelligence
- Automated market monitoring
- Efficient information filtering for analysts

DATASET OVERVIEW

Dataset: Twitter Financial News

- 21,107 financial tweets
- 20 financial topic categories
- Multi-class classification problem

Example categories:

• Analyst Update • Earnings • Markets • Stock Movement
• Macroeconomic News , etc...

Tweet Example:

1. “Tesla shares surge after strong earnings report”

- Label : Stock Movement

PROJECT WORKFLOW

Data Collection

Text Cleaning

Tokenization & Lemmatization

Feature Engineering

Model Training

Evaluation

Insight :
This structured workflow converts unstructured text into machine learning inputs.

NLP PIPELINE ARCHITECTURE

Raw Tweets

Text Cleaning

Tokenization

Vectorization

Deep Learning Model

Financial Topic Prediction

Insight :
This architecture enables automated financial text classification.

TEXT PREPROCESSING

Cleaning operations performed:

- Remove URLs
- Remove punctuation
- Convert to lowercase
- Remove stopwords
- Apply lemmatization

DATASET IMBALANCE ANALYSIS

Title: Label Distribution Analysis

Visual :
Bar Chart

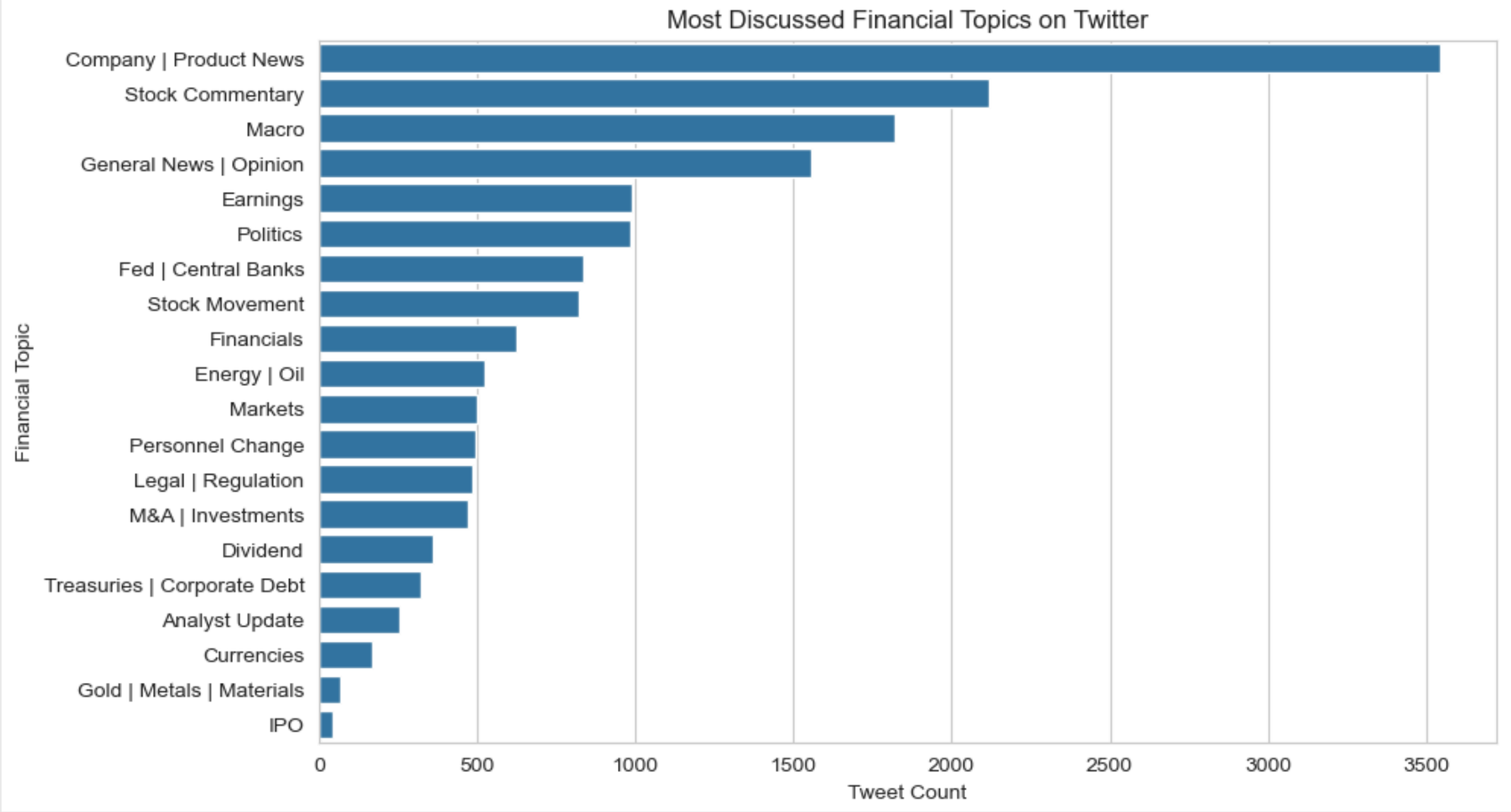
Insight :
Dataset is highly imbalanced.

Some categories like:

1. Markets
2. Company News

- appear significantly more frequently.
- Handling imbalance is critical for model accuracy.

LABEL DISTRIBUTION



TWEET LENGTH DISTRIBUTION

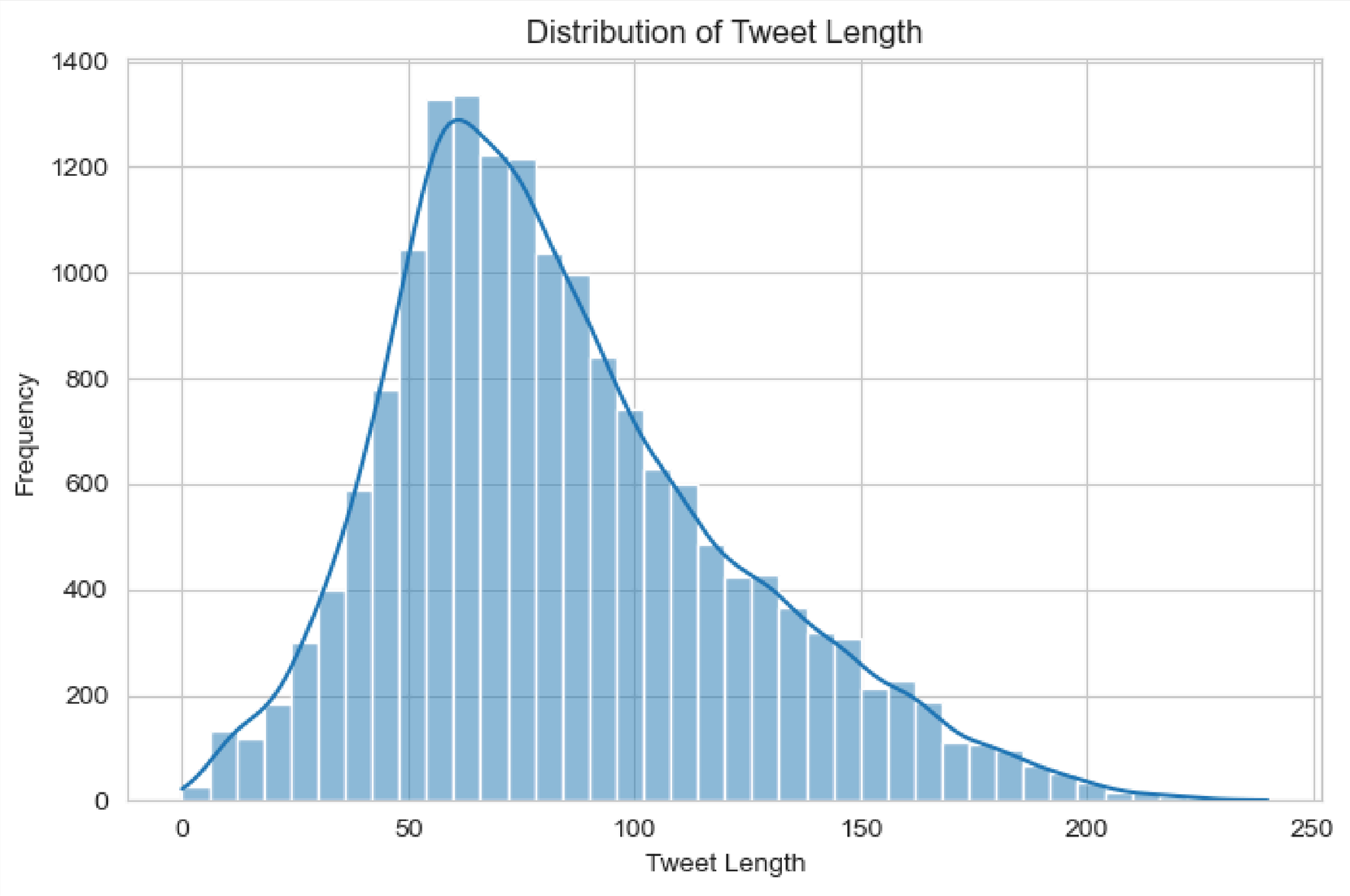
Title: Tweet Length Analysis

Visual :
Histogram

Insight :

- Most tweets are short textual messages, which confirms that models must handle short financial statements efficiently.
- This supports the use of tokenization and sequence-based NLP models.

LABEL DISTRIBUTION



WORD FREQUENCY ANALYSIS

Title : Financial News Word Cloud

Visual :
Word Cloud

Insight :
Large words indicate dominant financial themes such as:

- us
- new
- earnings
- stocks
- economy
- inflation

WORD FREQUENCY ANALYSIS

Most Frequent Words in Financial Tweets
new
say
second
quarter
company
year
inflation
china
report
global
share
stock
announce
result
sale
government
conference
call
cost
time
declares
dividend
higher
high
help
see
watch
plan
billion
trade
central
bank
cut
fund
back
increase
tesla
buy
russia
group
pre
deal
market
ceo
one
day
investor
week
continue
risk
may
june
according
interest
rate
holding
move
rise
fed
first
set
capital
said
energy
price
call
earnings
solution
recession
lower
demand
today
still
latest
month
data
europe
top
july
million
service
oil
growth
elon musk
support
industry
earnings
results
country
q earnings
partner
prime minister
ahead
amid
financial results
investment
companies
outlook
rate hike
hit
look
strong
ukraine
amazon
target
netflix
production
put
level
start
order
point
system
new york
contract
step
home
leader
acquire
platform
record
even
supply
potential
good
spy
999
future
ecb
fall
international
quarter earnings
close
take
expected
win
economy
stockmarket
wall street
american
launches
sign
consumer
right
another
agreement
appoint
wires wires
product
quarter financial
provide
expand
keep
decline
two
update
gaap eps
likely
make
chief
firm
european
short
world
end
est prev
call transcript
reuters
debt
much
key
use
value
face
next
former
open
expect
quarter results
technology
dollar
first time
uk regulatory
forecast
health
remain
expected
low
talk
euro
expectation
come
power
tsla
twitter
job
drop
profit
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uk
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eu
regulatory announcement
system
contract
step
home
new york
point
order
start
level
even
supply
put
production
face
next
value
make
likely
gain
eps
gaap
update
two
country
product
international
stockmarket
economy
twitter
tsla
technology
dollar
first time
power
expectation
euro
talk
low
expected
win
wall street
american
launches
sign
consumer
right
another
agreement
appoint
wires wires
quarter financial
provide
expand
keep
decline

SENTIMENT DISTRIBUTION

Title : Sentiment Distribution by Financial Topic

Visual :
Bar Chart

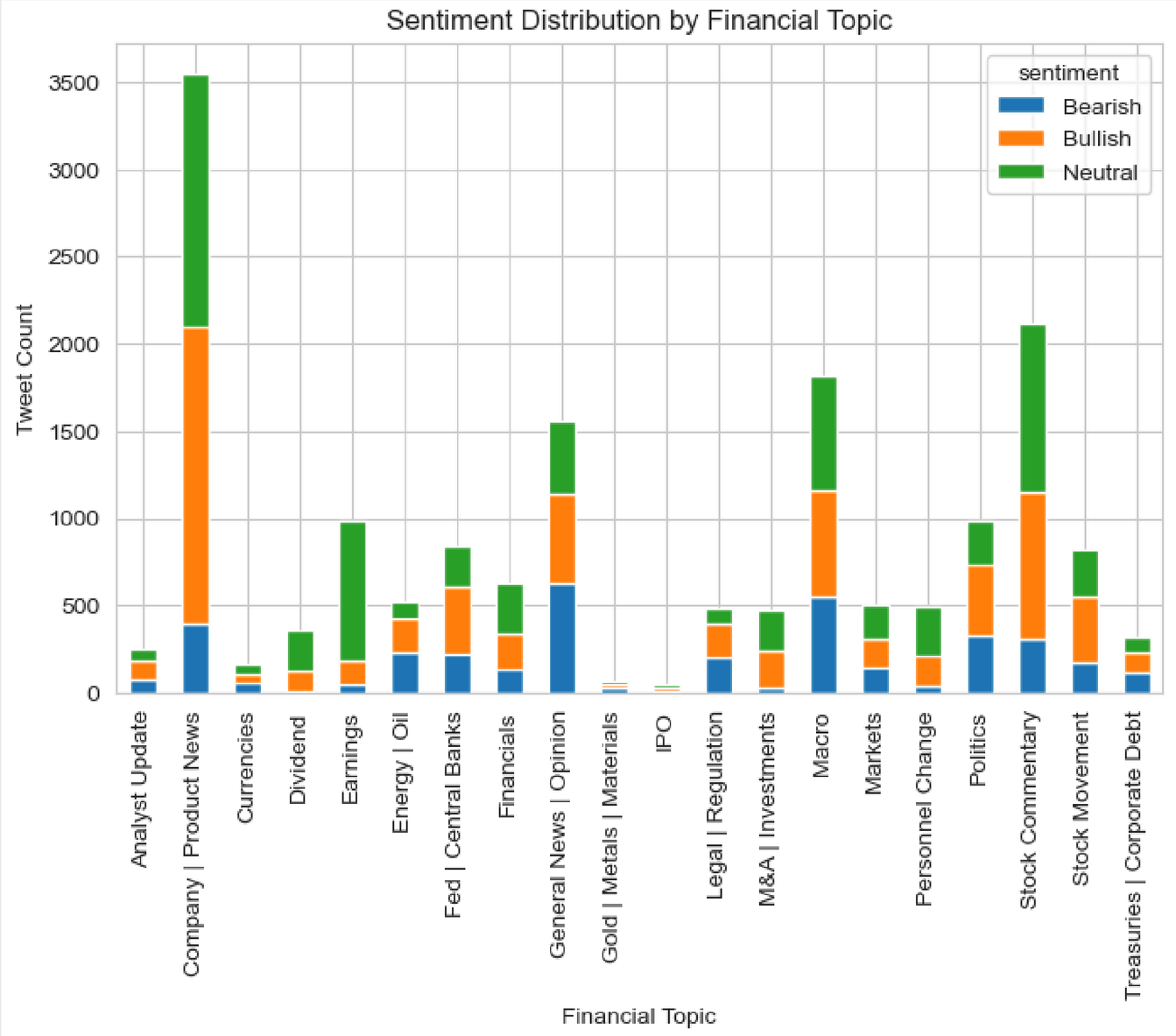
Insight :
The dataset shows different levels of:

- Bullish sentiment
- neutral sentiment
- Bearish sentiment

Understanding sentiment can help analyze market reactions and investor sentiment.

Business Question Answered
What sentiment patterns exist in financial tweets?

SENTIMENT DISTRIBUTION BY FINANCIAL TOPIC



FEATURE ENGINEERING

Business Question Answered :
How do we convert text data into machine learning features?

Insight :
Machine learning algorithms require numerical input, so text must be transformed using NLP techniques.

FEATURE ENGINEERING (DIAGRAM)

1. Financial Tweet
2. Text Cleaning
3. Tokenization
4. Vectorization
5. Machine Learning Model

FEATURE ENGINEERING

(TEXT VECTORIZATION)

Machine learning models require numerical input.
Methods used:

- TF-IDF
- Tokenization
- Sequence Padding

Example categories:
• Analyst Update • Earnings • Markets • Stock Movement
• Macroeconomic News , etc...

Tweet Example:

1. “Tesla shares surge after strong earnings report”

- Label : Stock Movement

TRAINING AND TESTING DATASET

To evaluate the model reliably, the dataset was divided into two parts:

- Training Data (file: train_data.csv) Used to train the machine learning model.
- Testing Data (file: valid_data.csv) Used to evaluate the model’s prediction performance.

This ensures the model learns patterns from historical data and is tested on unseen tweets.

MACHINE LEARNING MODEL SELECTION

For financial tweet classification, we used:
Logistic Regression

Reasons:

- Efficient for text classification
- Works well with TF-IDF features
- Provides interpretable results
- Fast training and prediction

Logistic regression serves as a strong baseline model for NLP classification tasks.

MODEL TRAINING PROCESS

WORKFLOW DIAGRAM

Tweets

TF-IDF Vectorization

Train-Test Split

Logistic Regression

Prediction

Insight :
This pipeline transforms unstructured financial text into structured machine learning inputs.

MODEL ACCURACY

Title : Model Performance Accuracy

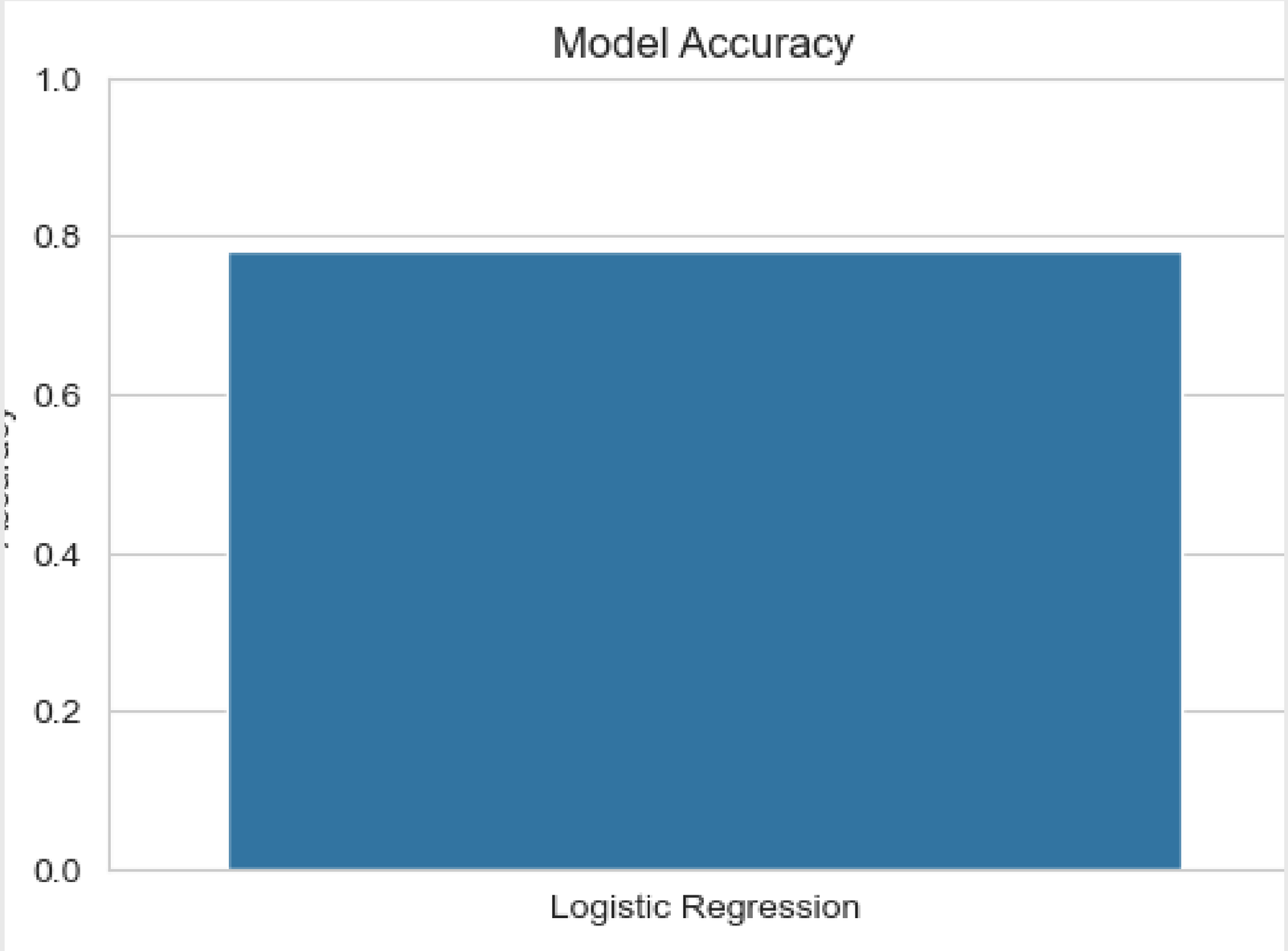
The trained model achieved strong classification performance.
Accuracy represents the percentage of tweets correctly classified into their financial topic.

Visual :
Bar Chart

Insight :
High accuracy demonstrates that financial tweets contain recognizable language patterns that machine learning can learn.

Business Question Answered
How accurately can the model classify financial tweets into the correct category?

MODEL PERFORMANCE ACCURACY



CONFUSION MATRIX

Title : Confusion Matrix – Prediction Analysis

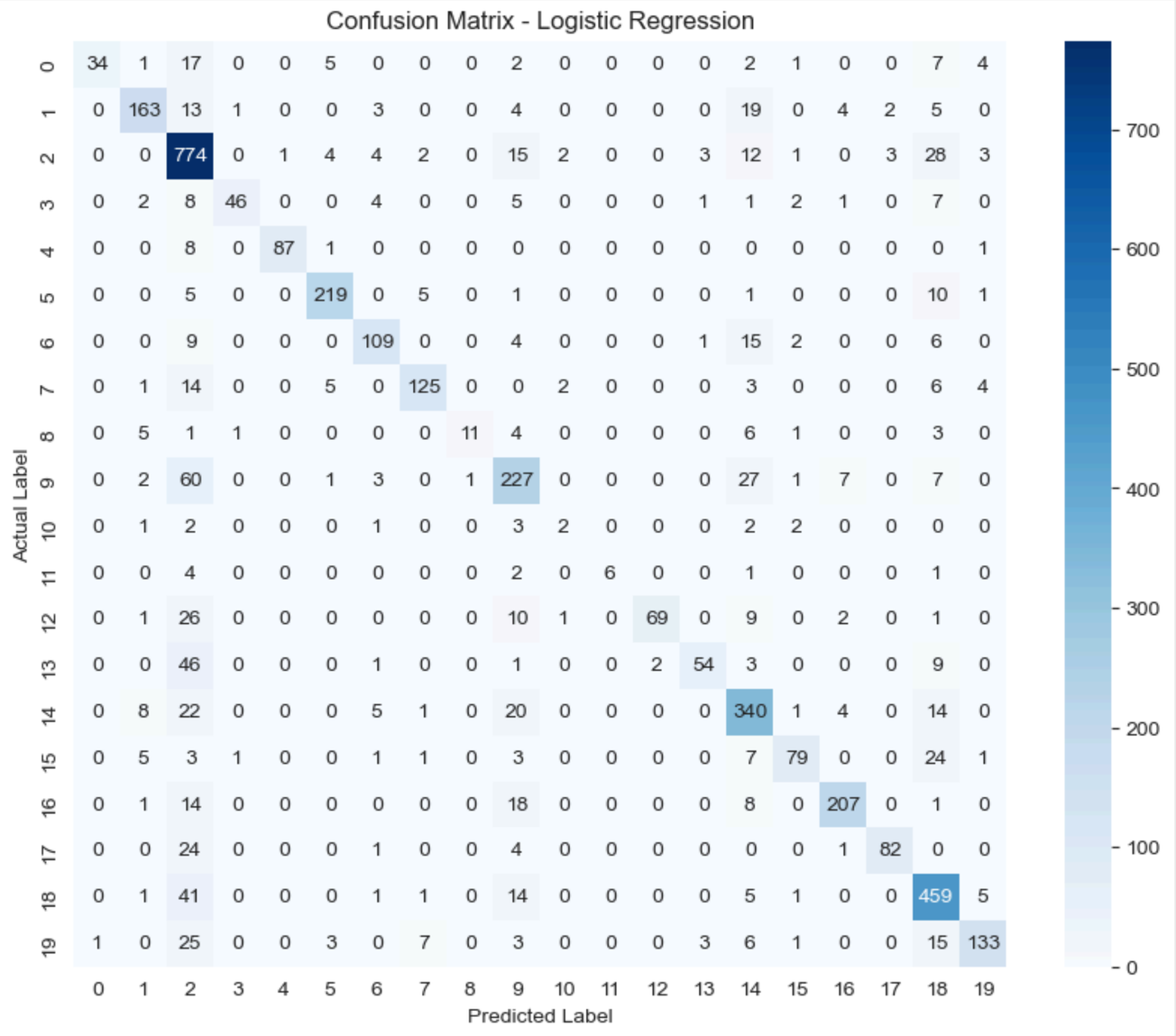
Visual :
Heatmap

Insight :

- Strong predictions appear along the diagonal, meaning the model correctly identifies many financial categories.
- Some confusion exists between similar topics such as market news and company updates.

Business Question Answered
Which financial categories are predicted correctly and where does the model make errors?

CONFUSION MATRIX – PREDICTION ANALYSIS



MODEL EVALUATION METRICS

Title : Model Evaluation Metrics

Visual :
Table from classification report
Metrics included:

- Precision
- Recall
- F1 Score

Insight :

- Balanced precision and recall indicate the model performs consistently across different financial categories.

Business Question Answered
Is the model reliable across all financial topics?

MODEL EVALUATION METRICS

	precision	recall	f1-score	support
0	0.97	0.47	0.63	73
1	0.85	0.76	0.80	214
2	0.69	0.91	0.79	852
3	0.94	0.60	0.73	77
4	0.99	0.90	0.94	97
5	0.92	0.90	0.91	242
6	0.82	0.75	0.78	146
7	0.88	0.78	0.83	160
8	0.92	0.34	0.50	32
9	0.67	0.68	0.67	336
10	0.29	0.15	0.20	13
11	1.00	0.43	0.60	14
12	0.97	0.58	0.73	119
13	0.87	0.47	0.61	116
14	0.73	0.82	0.77	415
15	0.86	0.63	0.73	125
16	0.92	0.83	0.87	249
17	0.94	0.73	0.82	112
18	0.76	0.87	0.81	528
19	0.88	0.68	0.76	197
accuracy			0.78	4117
macro avg	0.84	0.66	0.72	4117
weighted avg	0.80	0.78	0.78	4117

KEY INSIGHTS FROM DATA ANALYSIS

From analyzing financial tweets we discovered:

- Twitter contains valuable financial market information
- Financial tweets often focus on earnings, market movement, and company news
- Machine learning can successfully identify financial topics
- Financial social media data can be transformed into structured insights for investors and analysts.

REAL-WORLD APPLICATIONS

Potential applications of this system include:

- Financial news monitoring platforms
- Investment research tools
- Market sentiment analysis
- Algorithmic trading research
- Financial intelligence dashboards

PROJECT OUTCOME

This project successfully:

- Built a complete NLP classification pipeline
- Analyzed financial Twitter data
- Implemented machine learning classification
- Evaluated prediction performance
- Demonstrates how data science can convert unstructured financial text into actionable insights.

CONCLUSION

Financial markets produce large volumes of textual data every day.

This project demonstrates how Natural Language Processing and Machine Learning can automatically categorize financial tweets and support faster market analysis.

Automated financial text classification can significantly enhance investment research and financial intelligence systems.